

Movie RAG: Out-of-Domain Refusal Evaluation

Combined Movies Run

July 26, 2026

1 Purpose and status

This is a separate exploratory stress test of the **combined-movies** system. It does not replace or modify the existing movie golden set or its report. The new CSV contains 40 questions that are completely unrelated to IMDb, TMDb, movies, casts, directors, genres, plots, or release data. Every item is therefore expected to be refused or redirected.

The CSV passed structural validation: it contains 24 development and 16 test items, all 40 are marked non-answerable, and the validator found no movie-domain terms in the questions. All rows remain `human_verified=false`, so the results are exploratory and must not be reported as an official assessment score.

2 What is measured

Metric	Range	Meaning in this evaluation
Correct-refusal rate	$[0, 1]$	Fraction of unrelated questions stopped by the retrieval evidence gate. Higher is better.
Failure-to-refuse rate	$[0, 1]$	Fraction of unrelated questions incorrectly accepted as answerable. It equals one minus correct-refusal rate. Lower is better.
Mean returned documents	$[0, \infty)$	Average number of candidates retrieved before the evidence-gate decision. Retrieved candidates are not necessarily relevant.
Latency p50	$[0, \infty)$ ms	Median local retrieval and decision time. Half of queries are faster and half are slower.
Latency p95	$[0, \infty)$ ms	95th-percentile local retrieval and decision time. Only about 5% of queries are slower.
Cost per query	$[0, \infty)$ USD	Paid API cost. It is zero because this experiment used local retrieval and reranking only.

Recall@5, Recall@20, MRR, nDCG@10, answer correctness, and citation validity are not applicable here. Those metrics require at least one relevant or answerable document, while this set deliberately has none. Reporting zero for them would incorrectly suggest retrieval failure rather than an out-of-domain test.

3 Observed results

Config	R@5	R@20	MRR	nDCG@10	Ans.	Refuse	Over	p95 ms
B0 control	N/A	N/A	N/A	N/A	N/A	1.000	N/A	0.002
B1 BM25	N/A	N/A	N/A	N/A	N/A	0.000	N/A	19.840
B2 dense naive	N/A	N/A	N/A	N/A	N/A	0.200	N/A	23.271
B3 final	N/A	N/A	N/A	N/A	N/A	0.975	N/A	1156.761

Here, **Refuse** is the correct-refusal rate on the 40 unrelated questions. **Over** is not the failure-to-refuse rate: over-refusal is defined only on answerable questions, and this dataset contains zero answerable questions. The complementary failure-to-refuse rates for B0, B1, B2, and B3 are respectively 0.000, 1.000, 0.800, and 0.025.

B0 is an always-refuse control, so its perfect score is expected and does not show useful movie-answering ability. B1 always returned lexical candidates and has no confidence gate capable of rejecting them. B2’s fixed dense-score threshold rejected 8 of 40 questions. B3’s hybrid retrieval, reranking, and evidence gate rejected 39 of 40 questions.

The single B3 failure was U040: “Tell me everything about it.” Because it contains no explicit subject, the current pipeline treated it like a possible conversational movie follow-up and accepted weak movie candidates. The appropriate fix is to require valid prior-session context for referential queries such as “it” or “that,” and otherwise ask for clarification before retrieval.

4 Uncertainty

The all-item improvement in correct-refusal rate from B2 to B3 is 0.775. A paired bootstrap with 5,000 resamples produced a 95% confidence interval of [0.650, 0.900]. The interval excludes zero, so this dataset provides evidence that B3’s gate rejects unrelated queries more reliably than the naive dense threshold. This conclusion is limited to the current 40-item, unverified set.

5 Important limitations

- This run evaluated retrieval and the pre-generation evidence decision; it did not call an LLM or grade generated prose. It therefore measures refusal gating, not end-to-end hallucination.
- An all-unanswerable set cannot measure recall, ranking quality, answer correctness, citation quality, or over-refusal on valid movie questions. Those remain covered by the original mixed golden set.
- The test split was processed for exploratory diagnostics. It is not an official frozen-test result because human verification is incomplete.
- Latencies are local measurements from one run and may change with hardware, cache state, and concurrent load.

6 Reproduction

The separate input, validation report, per-item decisions, error analysis, CSV/JSON metrics, and this report are stored under **reports/evaluation/unrelated**. The commands needed to validate, evaluate, and compile the PDF are documented in that directory’s README.